

EE 446 TinyML Lab 8: ASL Gesture Recognition

Submission Report

1. Software Issue Fixed

The provided notebook used a softmax output for a two-class classification problem, but the model was compiled with mean squared error (`mse`) and mean absolute error (`mae`). The labels are one-hot encoded, so I changed the loss to `categorical_crossentropy` and the evaluation metric to `accuracy`.

```
model.compile(  
    optimizer='rmsprop',  
    loss='categorical_crossentropy',  
    metrics=['accuracy']  
)
```

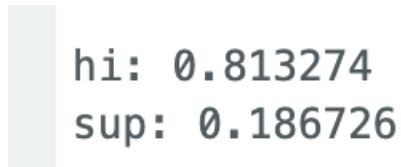
2. Consistency Changes

To keep the notebook consistent with the corrected training setup, I changed `METRIC_TO_PLOT` from `mae` to `accuracy`. The loss plots still use `loss` and `val_loss`. After training and evaluation, the model was converted to TensorFlow Lite and exported as `model.h` for Arduino deployment.

3. Arduino Test Results

The model was deployed on an Arduino Nano 33 BLE Sense Rev2 and tested with the Arduino Serial Monitor. Both gestures were correctly identified.

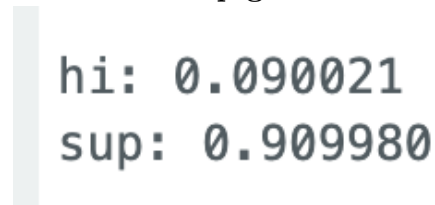
Hi gesture

A screenshot of an Arduino Serial Monitor showing the output of a gesture recognition test. The text displayed is "hi: 0.813274" on the first line and "sup: 0.186726" on the second line. The background is a light gray with a darker gray border on the left and top.

hi: 0.813274
sup: 0.186726

hi = 0.813274, sup = 0.186726

Sup gesture

A screenshot of an Arduino Serial Monitor showing the output of a gesture recognition test. The text displayed is "hi: 0.090021" on the first line and "sup: 0.909980" on the second line. The background is a light gray with a darker gray border on the left and top.

hi: 0.090021
sup: 0.909980

hi = 0.090021, sup = 0.909980

The first test was classified as `hi` with about 81.3% confidence. The second test was classified as `sup` with about 91.0% confidence.